5. React Libraries

▼ What is React?

- **React** is a JavaScript library used to build **user interfaces (UI)**, especially for **Single-Page Applications**.
- It focuses on building components.

> 💡 Think of React as the **brain** behind how your app behaves and how the components are structured.

- There are two main React libraries:
  1. react
  2. react-dom

▼ What is `main.jsx` in React?

- `main.jsx` is the **entry point of your React application**.
- It acts like the **starting engine** that connects React code to the real browser DOM.
- It is the file where:
  - The React app is initialized
  - The `App` component is rendered into the DOM
  - You connect React to the HTML file (`index.html`)

▼ What is `App.jsx` in React?

- `App.jsx` is the **root component** of your React application.
- It is the **main component** that acts as the **starting point** or **container** for all other components in your app.
- **Key Points About** `App.jsx`
  1. Root Component

- `App.jsx` is the top-level component.
- Every other component (like Header, Footer, Navbar, etc.) is usually nested inside `App.jsx`.
- It is the **first component loaded into the DOM** via `main.jsx`.

▼ What is `react` library?

- The **react** is a core library of React.
- The `react` package provides:
  - Tools to create components
  - Manage state and props
  - Handle lifecycle methods
  - Handle hooks (like `useState`, `useEffect`, etc)

▼ What is `react-dom` library?

- **ReactDOM** is the package that allows React to interact with the browser's **DOM (Document Object Model)**.

> 💡 If React is the brain, **ReactDOM is the bridge** between React and the browser.

- Responsible for rendering components into the UI/DOM tree.
- Imported via **import ReactDOM from** '`react-dom/client`'

▼ What does the `react-dom` library do?

- Renders your React components into the Real DOM.
- **Provide a method like**:
  - `ReactDOM.createRoot()` (for React 18+)

▼ `createRoot()` Method

- A function in React to create a root for rendering React components within a DOM node.

- It acts as a bridge between `index.html` and `main.js` files.
- It specifies where React code should render.
- Import via `import {createRoot} from "react-dom/client"`
- **Syntax**:

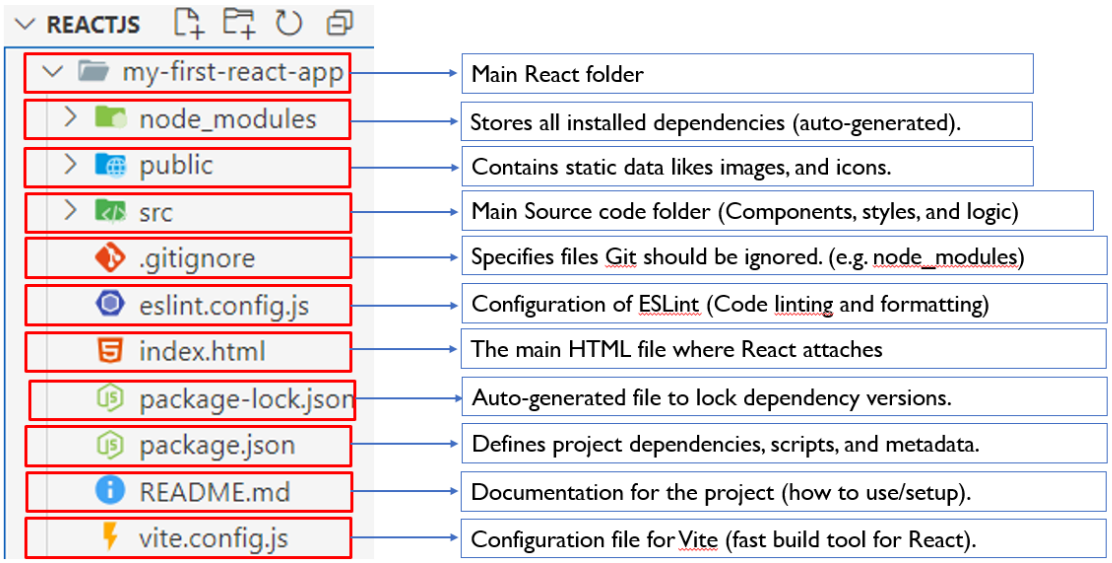
```
import { createRoot } from "react-dom/client";

createRoot(document.getElementById("root")).render(
  <h1>React 18 onwards use createRoot function</h1>
);
```

- **OR**

```
import ReactDOM from "react-dom/client";

ReactDOM.createRoot(document.getElementById("root")).render(
  <h1>React 18 onwards use createRoot function</h1>
);
```

▼ React Folder Structure

- Here's a **clean and scalable folder structure** for a **React + Vite** project.